

Table 9: (Appendix) RQ1 per-App diagnostics: denominator effect, continuous coupling, and evidence-set inflation. Each cell shows $w/o \rightarrow w/(\Delta)$.

App	$ V $ (abs)	$ V_{LCC} $ (abs)	$\bar{d}_j$ ↓	$Avg\Gamma(v)$ ↓	$Avg\Gamma(u) \cup \Gamma(v)$ ↓	$Avg\Gamma(u) \cap \Gamma(v)$ ↓
Slife	4052→3913 (-139)	2165→2248 (+83)	0.507→0.532 (+0.025)	1.472→1.523 (+0.051)	9.719→11.921 (+2.202)	1.121→1.138 (+0.016)
JayMe	2237→1924 (-313)	1321→1378 (+57)	0.525→0.610 (+0.085)	1.493→1.733 (+0.241)	8.827→14.143 (+5.316)	1.132→1.198 (+0.066)
Huawei Health	872→790 (-82)	271→393 (+122)	0.299→0.401 (+0.102)	1.212→1.334 (+0.122)	2.976→4.188 (+1.212)	1.062→1.098 (+0.036)
Hujiang English	1054→944 (-110)	493→580 (+87)	0.357→0.445 (+0.088)	1.308→1.457 (+0.148)	3.394→4.426 (+1.032)	1.088→1.127 (+0.039)
Tunni	7680→7340 (-340)	4345→4447 (+102)	0.531→0.561 (+0.030)	1.465→1.531 (+0.066)	20.280→20.802 (+0.522)	1.086→1.103 (+0.017)
IELTS Listening	898→805 (-93)	110→335 (+225)	0.294→0.402 (+0.108)	1.249→1.386 (+0.137)	2.495→3.187 (+0.692)	1.090→1.121 (+0.031)

Note. $Avg\Gamma(u) \cup \Gamma(v)$ and $Avg\Gamma(u) \cap \Gamma(v)$ are computed over eligible edges E' (endpoints u, v denote the cluster assignment of report r under a fixed boundary

Table 10: (Appendix) Supplementary comparison with generic external baselines for RQ2 under the same induced candidate pool and the same shared review-cost budget. Higher is better for nAUC@B, Cov@0.50B, Cov@0.75B, and Sel@B; lower is better for C/V.

Method	nAUC@B	Cov@0.50B	Cov@0.75B	Sel@B	C/V
ECFA	0.451	0.513	0.656	50.0	25.272
Louvain-Medoid	0.359	0.402	0.549	47.8	14.458
MMR	0.318	0.369	0.522	12.2	92.205
FacilityLocation	0.225	0.252	0.308	41.7	2.864

Note. Values are averaged across the 6 Apps. Louvain-Medoid is the strongest external competitor overall, but it still remains below ECFA on the main budget-coverage trade-off. MMR and FacilityLocation are substantially less competitive under the same protocol, suggesting that generic diversification/subset-selection objectives are poor fits for budget-aware functional-view selection in this setting.

A RQ1 Supplementary Results

This appendix provides additional diagnostics for RQ1. Table 9 reports per-App diagnostics used to (i) mitigate denominator-effect concerns by reporting $|V|$ and $|V_{LCC}|$, and (ii) quantify evidence-set inflation via provenance-set sizes.

B Additional RQ2 Details

B.1 Supplementary comparison with generic external baselines

To complement the controlled ablation in the main text, we additionally evaluate three representative generic external selection baselines under the same induced candidate pool and the same shared review-cost budget: **MMR** [3], **FacilityLocation** [24, 32], and **Louvain-Medoid**, which combines Louvain community detection [2] with medoid-style representative selection [20]. Among them, **Louvain-Medoid** is the strongest external competitor, but it still remains below ECFA in the overall budget-coverage-reviewability trade-off. By contrast, **MMR** and **FacilityLocation** are substantially less competitive under the same protocol. Overall, these supplementary results suggest that generic diversification/community-selection objectives are not strong substitutes for budget-aware functional-view selection in this setting.

For readability, the main paper discusses only **Louvain-Medoid**, while the complete results for all three external baselines are reported below.

C Additional RQ3 Details

C.1 LLM-assisted student annotation protocol for cross-cluster relevance

RQ3 does not rely on a purely automatic silver relevance signal. Instead, its main benchmark is constructed through an *LLM-assisted student annotation protocol* under fixed clustering boundaries.

Let $\mathcal{D}_A$ be the report corpus for an App A , and let $cluster_B(r)$ denote the cluster assignment of report r under a fixed boundary $B \in \{\text{TFIDF+HC, SBERT+HC, LLMCluster}\}$. For each query report $q \in \mathcal{D}_A$, RQ3 focuses only on the *cross-cluster* search space

$$\mathcal{U}_B(q) = \{r \in \mathcal{D}_A \setminus \{q\} \mid cluster_B(r) \neq cluster_B(q)\}.$$

Because exhaustively adjudicating all pairs in $\mathcal{U}_B(q)$ is impractical at scale, we construct for each query-boundary unit a manageable annotation pool directly from the raw reports under the fixed boundary. This preparation step is independent of ECFA: no ECFA extraction output, entity unification result, induced view, or retrieval ranking is shown to annotators or used as an annotation cue. For each candidate pair (q, r) with $r \in \mathcal{U}_B(q)$, an LLM is used only as an annotation assistant to generate a preliminary relevance suggestion and a short rationale for the current boundary-conditioned retrieval task.

Three student annotators then review each candidate pair together with the LLM-proposed suggestion and decide whether the pair should be accepted as *relevant* to the query under boundary B . The final label is determined by majority vote. We denote the resulting adjudicated cross-cluster relevant set for query q under boundary B by

$$Rel_B^{\text{cross}}(q) = \{r \in \mathcal{U}_B(q) \mid (q, r) \text{ is accepted as relevant by majority vote}\}.$$

The main RQ3 benchmark is therefore best understood as a *large-scale comparative retrieval benchmark over an adjudicated candidate universe*, rather than as a weakly supervised silver benchmark or an exhaustively labeled full-corpus benchmark. All main RQ3 metrics in the paper are computed with respect to these adjudicated labels under each fixed boundary. Methods are instantiated under their respective accessible retrieval scopes, but only adjudicated candidate pairs contribute labeled positives in metric computation. Accordingly, the main benchmark should be interpreted as a boundary-conditioned comparative evaluation within this adjudicated candidate universe.

Secondary blinded audit subset. To further examine robustness beyond the main benchmark, we additionally construct a smaller secondary blinded audit subset. We first sample 120 query reports in a stratified manner across the 6 Apps (20 per App), restricting sampling to queries that exhibit cross-cluster relevant evidence under all three fixed boundaries. Instantiating each sampled query under TFIDF+HC, SBERT+HC, and LLMCluster yields 360 query-boundary sample units in total.

For each sample unit, we form a blinded judgment pool by taking the union of the top-20 cross-cluster candidates returned by ECFA-Retr, Global, and MultiCluster-L5, and then removing duplicates. Three student annotators review each pooled candidate blind to method identity and rank position, and the final label is again determined by majority vote. Because this subset is much smaller but more tightly controlled than the main benchmark, we use it

as a robustness-oriented audit rather than as a replacement benchmark. On this subset, the paper reports Success@20, Recall@20, MRR@20, and nDCG@20 based on the resulting majority-vote labels. Recall-based metrics on this audit subset are computed with

respect to the adjudicated pooled candidate set for each sampled query-boundary unit, rather than with respect to the full corpus.